

Department of Computer Applications

THE MAHARAJA SAYAJIRAO UNIVERSITY OF BARODA

Bachelor of Computer Applications Division-B Semester – III
Programming with Java

Continuous Lab Assessment – II

Day and Date – Friday, 06/10/2023

Time – 01:00 Hour

Max Marks – 25

Note: Ensure that your program is organized into appropriate packages, utilizes inheritance and interfaces for code reusability and structure, properly encapsulates attributes with getter and setter methods, and handles exceptions appropriately. Document your code with documentation comments for clarity.

Que: Create a Java application for modeling a simple university system. The system should have different classes for students, professors, and courses. Implement the following classes, packages, and interfaces: (20 for Implementation + 5 for Comments and Documentation)

University Package:

1. Person Class (Abstract):

- Create an abstract class named Person with the following attributes and methods:
 - String name
 - int age
 - String address
 - String email
 - abstract void displayDetails() (to be implemented by subclasses)
- Provide appropriate getter and setter methods to encapsulate the attributes.
- Choose the appropriate modifiers for the attributes.
- Implement a parameterized constructor to initialize the attributes.

2. Student Class (Inherits from Person):

- Create a Student class that inherits from the Person class.
- Add the following attributes:
 - int studentId
 - String major
 - float gpa
- Provide appropriate getter and setter methods to encapsulate the attributes.
- Implement the displayDetails method to display the student's details.

3. Professor Class (Inherits from Person):

- Create a Professor class that inherits from the Person class.
- Add the following attributes:
 - int employeeId
 - String department
 - float salary
- Provide appropriate getter and setter methods to encapsulate the attributes.
- Implement the displayDetails method to display the professor's details.

Course Package:

4. Course Interface:

- Create an interface named Course with the following methods:
 - void enrollStudent(Student student) throws EnrollmentException
 - void listStudents()

5. CourseImpl Class (Implements Course):

- Create a class named CourseImpl that implements the Course interface.
- Add an attribute:
 - String courseName
- Implement the methods to enroll students in the course and list enrolled students.
- Implement exception handling for the enrollStudent method. Create a custom exception EnrollmentException that is thrown when a course is full (e.g., maximum student count is reached).

Main Package:

6. UniversityTest Class (Main Class):

- Create a UniversityTest class with a main method for testing your university system.
- In the main method, create instances of students, professors, and courses.
- Enroll students in courses, and handle exceptions if the course is full.
- Display student and professor details.
- List students in courses.

Documentation:

7. Generate the Documentation of this application using JavaDoc Utility.